

Support Ticket Department Routing — Model Deployment Recommendation

Alifya Saify, Mrunali Dhakate, Roopa Shankar Rao, May 2026

BOTTOM LINE We recommend TechNova proceed with a pilot deployment of the automated ticket routing system. The model is expected to significantly reduce misrouting, shorten resolution times, and improve overall customer satisfaction. Crucially, the system has been specifically calibrated to ensure reliable detection of high-risk security issues.

BUSINESS IMPACT

Metric	Current State	With Model (Estimated)	Impact
Resolution time (misrouted)	14.8 hours	4–6 hours	↓ Faster by ~60–70%
Customer satisfaction	Baseline	+22 points	↑ Improved experience
Annual labor cost	High (manual routing)	Reduced	≈ \$1.2M savings
Routing accuracy	Manual, inconsistent	≈ 85%+	↓ Fewer reroutes

THE APPROACH Our solution acts as a smart dispatcher. It reads the text of each incoming support ticket and combines it with customer account details (such as account age, priority level, and subscription usage). By learning from past tickets, it recognizes patterns to predict the correct department. We found that the actual text of the ticket is the strongest predictor but combining it with account details helps the system make the right call when a request is brief or vague.

To protect the business, we built in a "better safe than sorry" rule for Security tickets: if a ticket shows even a moderate chance of being a security incident, it is routed directly to the Security team. This guarantees we catch over 90% of critical security issues immediately.

KEY RISKS

- **Security False Positives:** Because the system is highly cautious about security risks, the Security team will receive some benign tickets. This is an intentional trade-off to ensure no critical threats are missed.
- **Evolving Terminology (Data Drift):** If customers start using new terminology or if new product features are introduced, the system's accuracy may gradually decrease without periodic updates.
- **Adoption Risk:** Staff may hesitate to trust automated routing initially, requiring effective change management and clear communication during the pilot phase.

RECOMMENDATION Run a controlled 90-day pilot deployment. During this phase, closely track the routing accuracy—especially for the Security and Onboarding departments—and measure the actual improvements in ticket resolution times. If the business metrics align with our testing results, proceed to a full-scale operational rollout.